POLARIZATION DETECTION, POLARIZATION TYPE CLASSIFICATION, AND POLARIZATION MANIFESTATION IDENTIFICATION

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GROUP-2

POLAR DATASET(NLP)

Team Members

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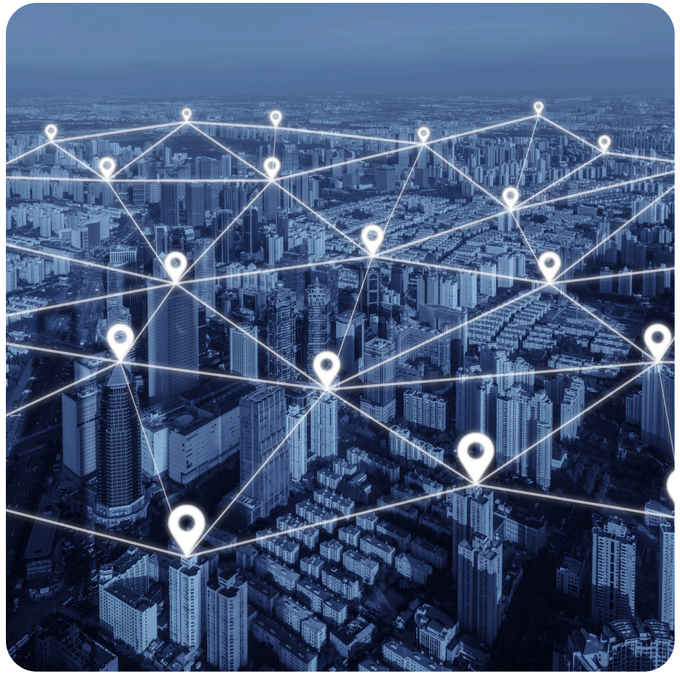
POLARIZATION DETECTION AND CLASSIFICATION

600
Text records

Dataset Overview

16
Original CSV columns

- Dataset: PunKhattgar_PolarAnnotations_Assignment2.csv
- Source: YouTube/Facebook comments and Formal Texts from Wikipedia in Myanmar language
- Tasks: Polarization Detection, Type Classification, and Manifestation Identification



Project Tasks

TASK 1: Detection

- Polarized / Not Polarized

TASK 2: Type Classification

- Political
- Racial/Ethnic
- Religious
- Gender/Sexual
- Other

TASK 3: Manifestation Identification

- Stereotype
- Vilification
- Dehumanization
- Extreme Language
- Lack of Empathy
- Invalidation

600
Text records

16
Original CSV columns

Data Preparation

Data Preprocessing

- Load the CSV dataset.
- Check missing values and duplicates.
- Prepare metadata.
- Create separate datasets for Tasks 1, 2, and 3.
- Split data into Train, Validation, and Test sets.

Data Split

- Training: 420 (70%)
- Validation: 90 (15%)
- Testing: 90 (15%)



MODEL Workflow

Dataset → Data Preprocessing → TF-IDF Feature Extraction → Model Training → Evaluation

MODEL PERFORMANCE COMPARISON

Task	Model	Result
Task 1	Logistic Regression	Test Accuracy: 83.33%
Task 1	Linear SVM	Validation F1: 0.21
Task 2	One-vs-Rest Logistic Regression	Macro-F1: 0.3205
Task 3	One-vs-Rest Logistic Regression	Macro-F1: 0.2711

Task-1: Polarization Detection Model

```
============================================================
TASK 1 — BINARY POLARIZATION CLASSIFICATION
============================================================
Test labels: 90
Predictions: 90
Accuracy : 0.8333
Precision: 0.2
Recall   : 0.2222
F1-score : 0.2105

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.90      0.91        81
           1       0.20      0.22      0.21         9

    accuracy                           0.83        90
   macro avg       0.56      0.56      0.56        90
weighted avg       0.84      0.83      0.84        90
```

- Test Accuracy: 83.33%
- Precision: 20.00%
- Recall: 22.22%
- F1-score: 21.05%

Task-2: Polarization Type Classification

```
============================================================
TASK 2 — POLARIZATION TYPE CLASSIFICATION
============================================================
Actual labels: (9, 5)
Predictions  : (9, 5)
IDs aligned: True
Macro-F1: 0.3205
Micro-F1: 0.3333
Macro Precision: 0.3667
Macro Recall: 0.3167

Classification Report:
               precision    recall  f1-score   support

    political       0.25      0.33      0.29         3
racial/ethnic       0.25      0.25      0.25         4
    religious       0.00      0.00      0.00         1
gender/sexual       1.00      0.50      0.67         2
        other       0.33      0.50      0.40         2

    micro avg       0.33      0.33      0.33        12
    macro avg       0.37      0.32      0.32        12
 weighted avg       0.37      0.33      0.33        12
  samples avg       0.22      0.28      0.24        12
```

- Macro-F1: 0.3205
- Micro-F1: 0.3333
- Exact-match accuracy: 11.11%

Task-3: Polarization Strategy Classification

```
============================================================
TASK 3 — POLARIZATION STRATEGY CLASSIFICATION
============================================================
Actual labels: (9, 6)
Predictions  : (9, 6)
IDs aligned: True
Macro-F1: 0.2711
Micro-F1: 0.5789
Macro Precision: 0.2917
Macro Recall: 0.2708

Classification Report:
                  precision    recall  f1-score   support

      stereotype       1.00      0.62      0.77         8
     vilification       0.75      1.00      0.86         6
  dehumanization       0.00      0.00      0.00         2
extreme_language       0.00      0.00      0.00         1
 lack_of_empathy       0.00      0.00      0.00         1
    invalidation       0.00      0.00      0.00         2

       micro avg       0.61      0.55      0.58        20
       macro avg       0.29      0.27      0.27        20
    weighted avg       0.62      0.55      0.56        20
     samples avg       0.63      0.51      0.51        20
```

- Macro-F1: 0.2711
- Micro-F1: 0.5789
- Exact-match accuracy: 11.11%
- Hamming loss: 0.2963

Presented by Pun Khattgar

THANK YOU